

Real-time Holographic Media System Utilizing HBM-based Holography Processor

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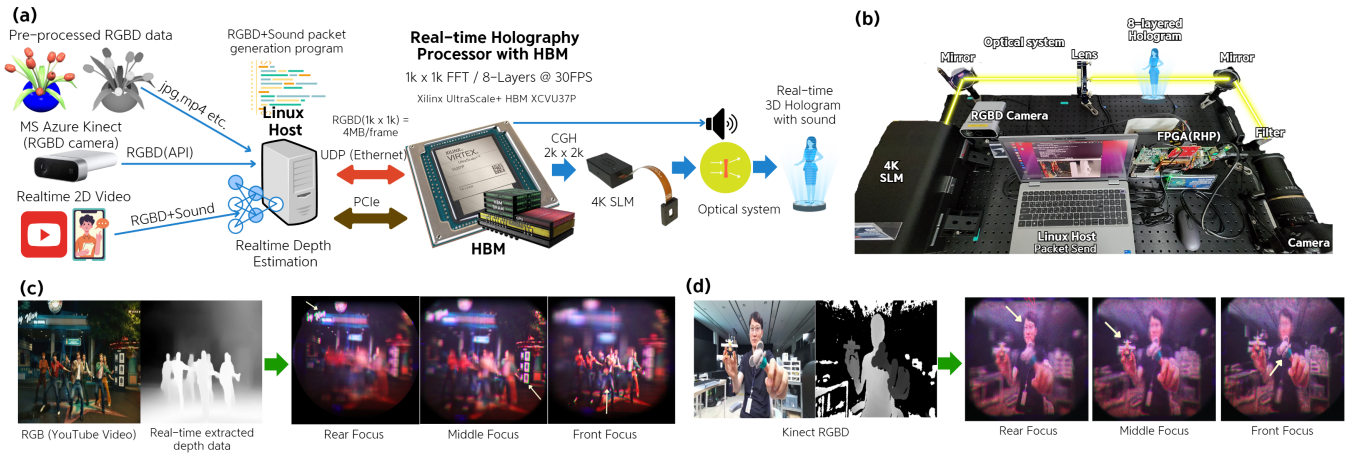


Figure 1: Real-time holographic media system: (a) conceptual diagram, (b) photograph of implemented demonstration system, experimental results of optically reconstructed holographic movies from input of (c) real-time 2D video, and (d) RGBD camera.

Abstract

This paper introduces a real-time holographic media system that converts 2D or RGBD videos into 3D holograms. The core of this system includes a Linux host that extracts depth information from 2D images and transmits it via packets, and a holography processor leveraging high-bandwidth memory (HBM) to generate volumetric holograms with 8-layer depths at 30 frames per second (fps). Demonstrations highlight the potential of the proposed system for advancing future holographic media technologies.

ACM Reference Format:

Wonok Kwon, Sanghoon Cheon, Kihong Choi, and Keehoon Hong. 2024. Real-time Holographic Media System Utilizing HBM-based Holography

Processor. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3681756.3697871>

1 Introduction

Computer-generated holograms (CGH) use point source-, polygon-, and layer-based algorithms, and as the number of these increases, the representation of depth information becomes richer, but the computational load also increases, making real-time playback difficult [Park 2017]. Using dedicated hardware with a field-programmable gate array (FPGA) to calculate CGHs addresses this limitation with high performance and efficiency, along with the benefits of low power consumption and miniaturization, making it suitable for applications like automotive head-up displays. Samsung reported a representative example of hologram generation hardware utilizing an FPGA, which employs the layer-based CGH algorithm requiring input images of color and depth information [Kim et al. 2019]. They used DDR4 memory to store 2D-FFT data and perform real-time CGH processing through two memory channels. Despite the

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SA Posters '24, December 03–06, 2024, Tokyo, Japan
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ACM ISBN 979-8-4007-1138-1/24/12
<https://doi.org/10.1145/3681756.3697871>

disadvantage of reduced hologram quality, they performed column-wise FFT on only 25% of the data from the row-wise FFT results to compensate for the slow column-wise FFT processing speed. In this study, we propose an FPGA-based CGH processor utilizing multi-channel high-speed HBM memory and efficient data processing algorithm to realize real-time holographic media system. The latency of the designed holographic media processor has been reduced to 28 ms, allowing for real-time data processing without any loss of hologram quality and bandwidth. We also developed a demonstration system capable of displaying a full-color volumetric hologram with 8-layer depth information at 30 fps using a 4K resolution spatial light modulator (SLM).

2 Real-time holographic media system

The real-time holographic media system consists of a packet generation program, a core CGH calculation processing unit named real-time holography processor (RHP), an SLM, and optics for displaying holographic images or movies, as shown in Fig. 1(a).

2.1 Packet generation program

A packet generation and transmission program running on a Linux host converts the RGB image, depthmap, and additional sound information (RGBDS) into packets and passes them to the RHP via the PCIe or Ethernet interface. The packet generation program transmits $1K \times 1K$ RGBDS data (4MB) at a rate of 30 fps. Various types of input can be used in the proposed system such as a pre-processed RGBD format videos or images, real-time RGBD data from color-depth cameras, and RGBDS format. Notably, the RGBDS can be generated in real-time as the form of packets from typical 2D videos, without depth data, using a real-time depth estimation deep learning model [Birkel et al. 2023].

2.2 Real-time holography media processor

We have developed a single-layer (no depth information implemented), monochromatic (color implemented using a time-division multiplexing), on-axis real-time holography processor using a simplified Fresnel propagation using a single FFT [Kwon and Cheon 2023]. In this study, we developed a general-purpose CGH processor that can generate off-axis color holograms with 8-layer depths in real-time using high-speed HBM memory. The proposed system leverages the advantages of high-speed HBM with a large number of channels and applies double buffering to reduce the column-wise FFT data read time. Additionally, an off-axis algorithm is added to the RHP architecture to remove the DC and conjugate noise from the reconstructed hologram. The hardware architectures for the RHP and HBM memory hardware structures are shown in Fig. 2. The RHP core block consists of various user hardware for generating CGH and has an AXI interface for a total of 9 channels of HBM memory. There are 6 and 3 channel HBM interfaces for 2D FFT and pre/post-processing, respectively. In the HBM memory architecture, HBM 0 to 5 are used to store row FFT results and have a double buffering structure by allocating two channels per RGB channel to compensate differences in processing speed between row and column cores. Row FFT operates efficiently because data is written to contiguous addresses, but column reads for column FFT are slow because data is read from discontinuous addresses. To

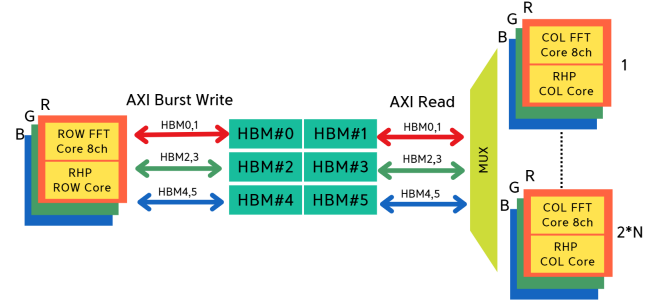


Figure 2: HBM memory architecture of the RHP to compensate for slow column read processing.

improve column processing speed, the number of column hardware logic unit is expanded to double ($N=1$). The FFT size used in the RHP core is $1K \times 1K$. The row and column core uses 24 ($RGB \times 8$ -layers) 48 ($RGB \times 8$ -layers $\times$ dual processing) FFT cores, respectively. Therefore, the RHP core uses a total of 72 $1K \times 1K$ FFT cores.

2.3 Optical system for holographic display

The viewing window-type holographic display optics are built to display the resultant holograms. A 3-inch lens with a focal length of 500 mm was used, and a viewing window of 13.8 mm was obtained at the viewing position. The hologram videos generated by RHP system in real-time are optically reconstructed at a distance of 1.25 m in front of the SLM, with the volume thickness of 400 mm.

3 Results and conclusion

Figure 1(b) shows a photograph of the experimental setup for the proposed system. The proposed real-time holographic media system was implemented and demonstrated on the Xilinx UltraScale+ HBM VCU128 FPGA evaluation board. In the experiment, a 2D Youtube video is converted into an off-axis hologram in real-time. As a resultant video (<https://youtu.be/qnGKDgL2rds>), the 2D video and estimated depthmap are successfully played back as 4K resolution hologram with 8-layer depths with sound at a speed of 30 fps. The RGB/depth of the input packet and the output hologram observed at the viewing window position are displayed through the input image and the monitor, respectively. The experimental results for different types of input images are shown in Figs. 1(c) and (d). In each experiment, it can be observed that the layer in which the camera is focused is sharp, while the other layers are blurred.

In this paper, we presented a real-time holographic media system capable of generating 4K resolution, 8-layer depth holograms at a speed of 30 fps using 2D videos without depth information as input, in addition to the commonly used RGBD video format. To achieve this, high-speed HBM and double buffering techniques are applied to reduce the latency occurring during hologram calculations. Furthermore, since the proposed system can convert any 2D video into holograms in real-time with low power consumption and small form factor, it can be applicable to various applications such as holographic telepresence, holographic video streaming, and automotive holographic head-up displays.

Acknowledgments

This work was supported by Institute for Information and Communications Technology Promotion (2019-0-00001).

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